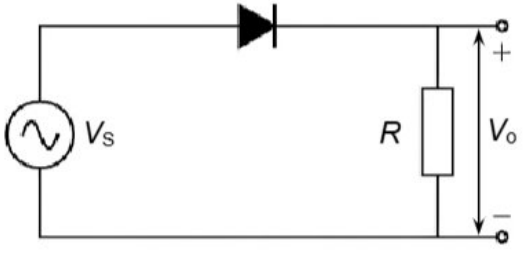
18 An ideal semiconductor diode is connected in series with a load resistor $R = 50\,\Omega$ and an alternating voltage source $V_S = 12.0\sin(100\pi t)$. The output voltage V_o is measured across the load resistor R.



What is the average power dissipated in the load resistor R over one complete cycle?

A 0 W **B** 0.72 W **C** 1.44 W **D** 2.88 W